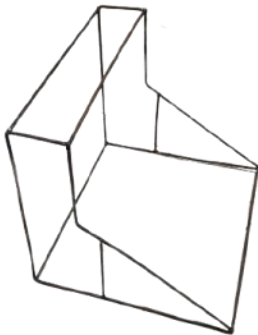


THE MASSIVE GRAVITY DAM



What?

- Design and construct a dam that stores **50L+** of water in its container and withstands **10Kg+** weight on its body.
- Performed a **metallic maquette** to **analyze** the final requirements.

How?

- Used **sweeter cement**, **gallstone** and **iron rods** with fixing bonds.
- Designed a **wooden mold** to pour the **concrete** inside it and build the body of the **gravity dam**.

Results

- The gravity dam fulfilled its purpose (stored **50L+** and withstand **10Kg+**) and earned an **A** grade.

BLUE ENERGY GENERATOR



What?

- Reduce the reliance on **non-renewable energy** usage by using **hydroelectric generators** and **dynamos** for **rural cities**.

How?

- Used a **DC motor**, **plastic tubes** and **batteries** to **store** the **energy** outputs.

Results

- Produced **156.57 Joules**, **6.14 Volts** and **0.085 ampere**, which meets the design requirements.

THE FABRIC FILTER



What?

- Reduce **air pollution** by developing a **two-stage** air filtration system to be used in **factories** and **chimneys**.

How?

- The project used **Arduino Uno**, **MQ-135 sensor** for detection and other **chemicals** for filtration.

Results

- The after testing analysis reflects the **success** of filtering **51.7%** of pollutants in **7 minutes**.

ECO CYCLE: A SUSTAINABLE WASTEWATER FILTRATION SYSTEM

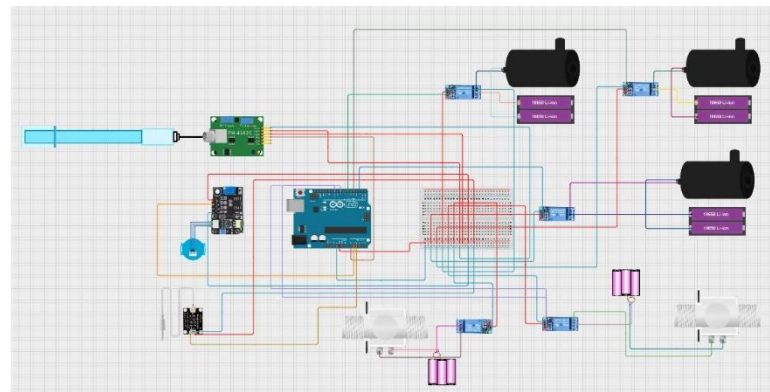


What?

- Reduce **water pollution** by developing a **three-stage** water filtration system to be used in **factories**.
- Used **physical stage**, **chemical stage**, and another **chemical stage**.

How?

- The project used **Arduino Uno**, **pH**, **TDS** and **Turbidity sensors** for detection and other **chemicals** for filtration.



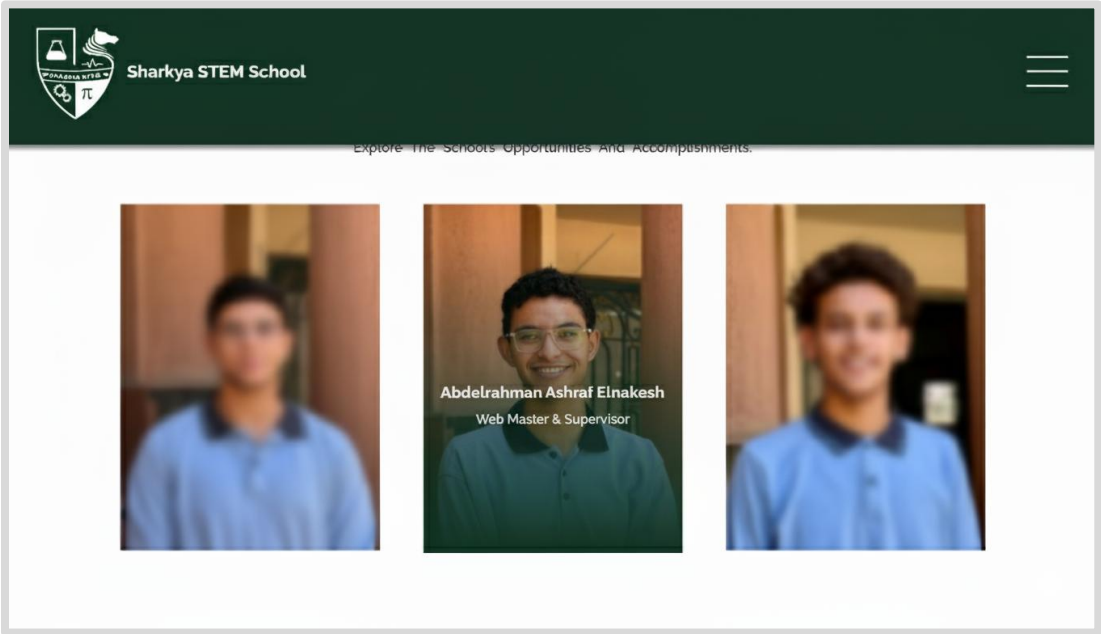
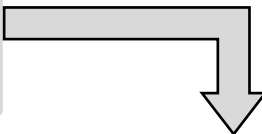
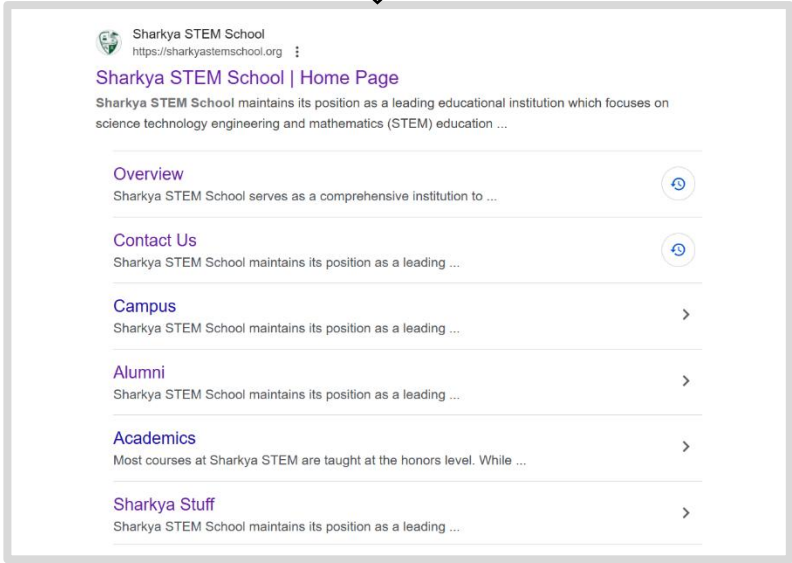
Results

- The after testing analysis reflects the **success** of filtering water from all kinds of pollutants in **3 cycles**.

SHARKYA STEM SCHOOL WEBSITE

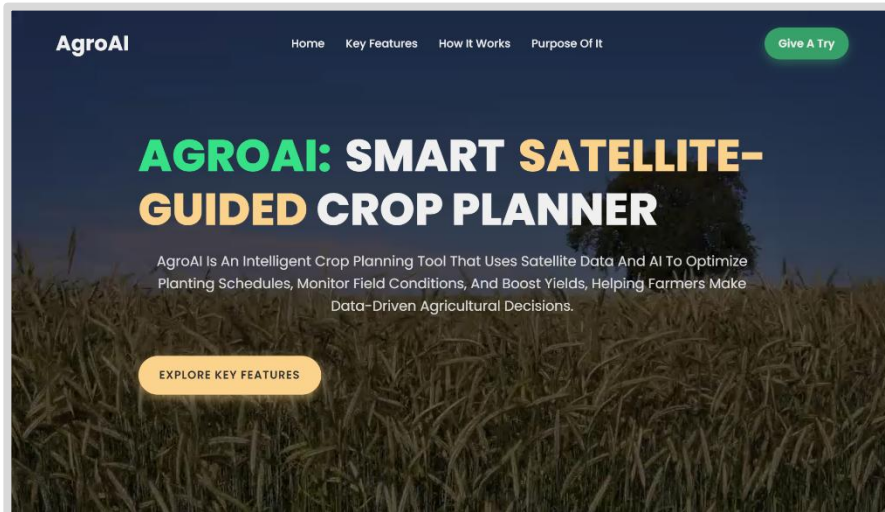
sharkyastemschool.org

Name	Date modified	Type	Size
css	7/30/2025 1:57 AM	File folder	
documents	7/30/2025 1:57 AM	File folder	
fonts	7/30/2025 1:57 AM	File folder	
html	7/30/2025 1:57 AM	File folder	
comps	7/30/2025 1:57 AM	JavaScript Source File	2 KB
footer_external_responsive	7/30/2025 1:57 AM	CSS Source File	4 KB
HomePage	7/30/2025 1:57 AM	CSS Source File	20 KB
index	7/30/2025 1:57 AM	Microsoft Edge HTM...	13 KB
main	7/30/2025 1:57 AM	CSS Source File	8 KB
navbar	7/30/2025 1:57 AM	JavaScript Source File	4 KB
navbar_externalFile	7/30/2025 1:57 AM	CSS Source File	10 KB
script	7/30/2025 1:57 AM	JavaScript Source File	4 KB
script_2	7/30/2025 1:57 AM	JavaScript Source File	1 KB

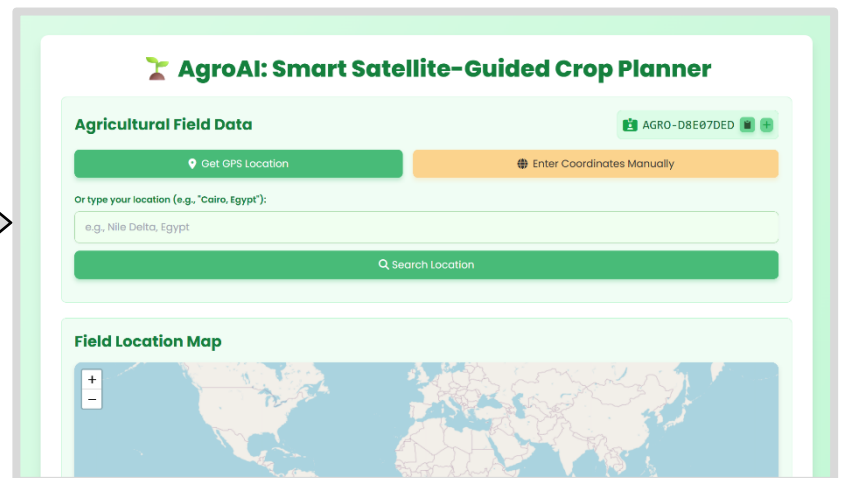
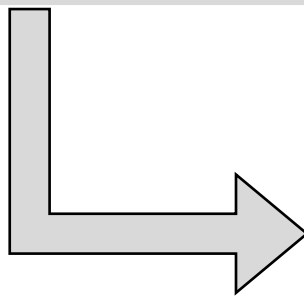


- Started developing the school's **first** official website in **October 2024** using **HTML, CSS, and JS** while using **databases** to manage **Google Site Maps**.
- Note:** Coded **3k+** lines, collected **10k+** impressions and **250+** records.
- Launched the school's **first** official website in **June 2025** using **Hostinger domains** while using **databases** to manage **Google Site Maps**.
- Note:** Published the website which includes **192** files and **6** folders; uploading **230 MB**.

AGRO AI: SMART SATELLITE GUIDED CROP PLANNER

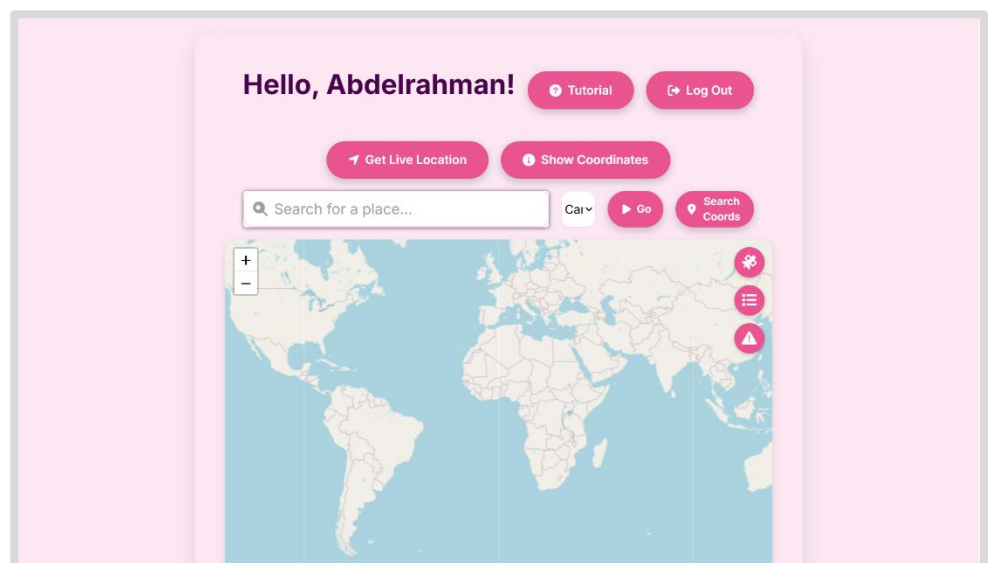


- Coded **AgroAI** for assisting **farmers** and **agriculturally based companies** in farming new lands and face **desertification** and **drought**.
- Providing a quick **revenue** based on available resources for farming efficiently and saving money.

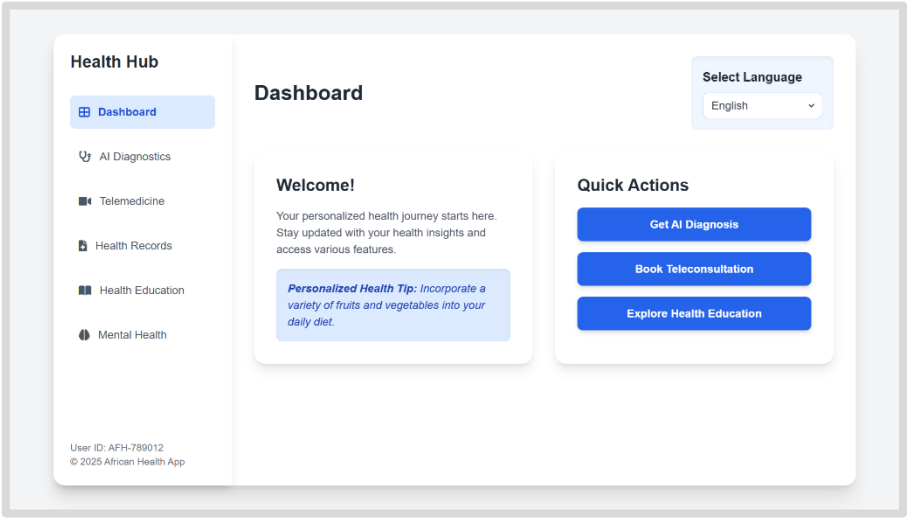


HER GUARDIAN

- Developed **HerGuardian** for assisting **women**, **teenagers** and **girls** in being safe while going outside their homes.
- Solving the SDG-5 (Gender Equality) using **HTML**, **CSS**, and **JS**.

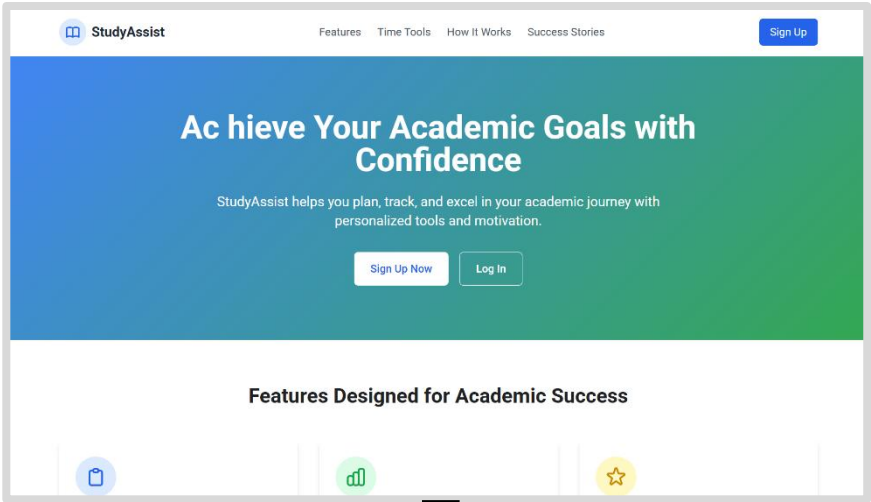


AFRICARE CONNECT

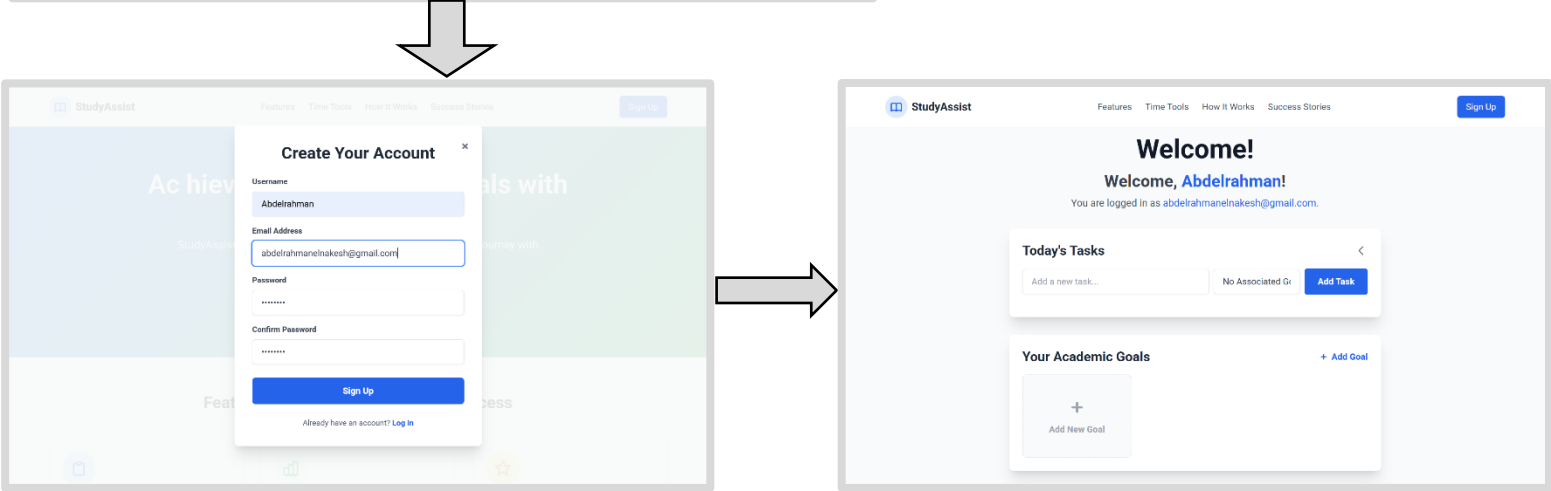


- Computed **AfriCare Connect** for assisting **African** poor people in booking **hospital** and **doctors’** medical examinations.
- Translated the application into **Arabic, French** and **English** to spread easily and ease the usage for all **Africans**.

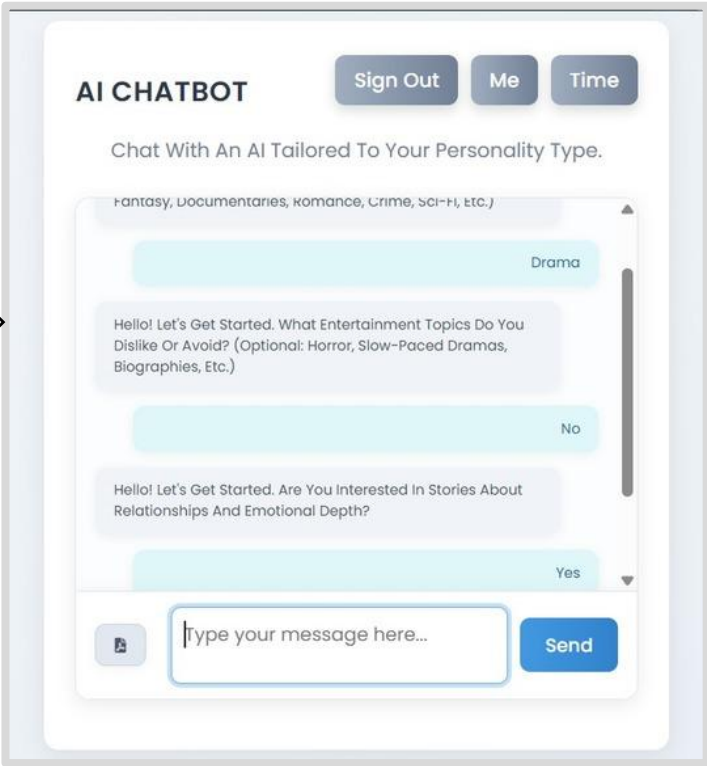
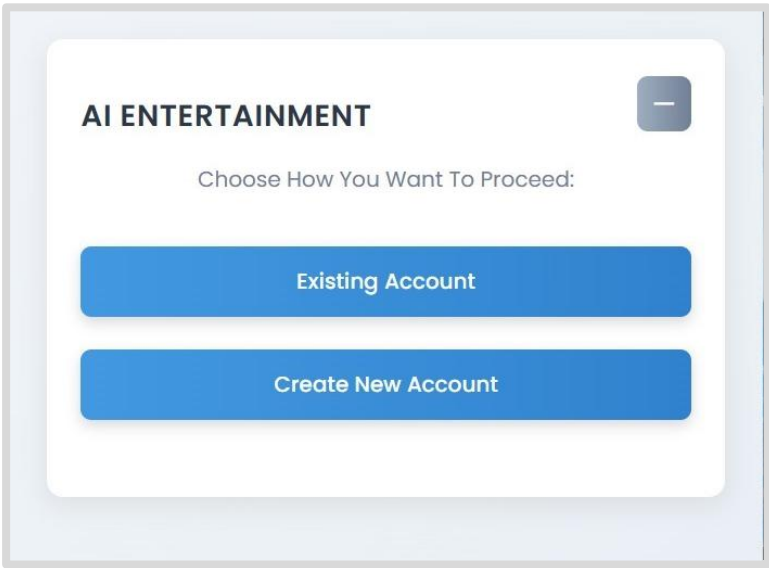
STUDYASSIST



- Programmed **StudyAssist** for improving the productivity of **students** and motivating them and managing their time effectively.
- Used **images** and **designs** to attract users and keep them **entertained** while studying and working.



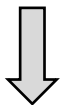
PERSONALITY AWARE ENTERTAINMENT AI



- Constructed **Personality Aware Entertainment AI Google Extension** for improving the entertainment experience for all kinds of personalities available.
- Used some easy entertaining **questions** to determine the personality and it's favorites using **AI** and **AI Chatbots**.

SMART IN-VEHICLE ALCOHOL DETECTION & SAFETY SYSTEM

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5554738



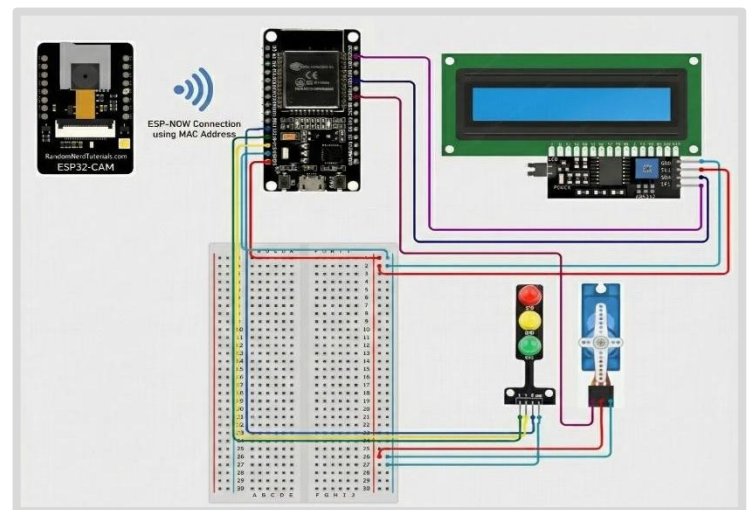
- Published a **research paper** about in-vehicle alcohol detection and advanced safety systems and designed a **scientific poster** about the **publication**.

FLOWGUARD: SMART CAMERA TRAFFIC MANAGEMENT



- Developed a smart Artificial Intelligence (AI) traffic management system.
- Applied an **ESP-NOW** wireless communication protocol between an **ESP-32 CAM** and an **ESP32 Development Board**.
- Used an **LCD I2C** screen for representing the information and instructions.

- Connected a mini-servo motor, traffic light kit, and an LCD I2C (16x2) with an **ESP32 Development Board**.
- Programmed the actions of them all with the commands of another **ESP32-CAM** using **wireless** communication.
- The Model was uploaded on the **ESP32-CAM CH340 board** for the detection purposes.



- Trained, tested, and analyzed **343 images** of the objects (Cars and an Emergency Ambulance Car); achieving an efficiency of **98.3%**.

LOGOS & FURTHER INFORMATION



School's website Logo



AgroAI project Logo



HerGuardian project Logo



AfriCare Connect project Logo



StudyAssist project Logo



Personality AI project Logo

Note!

All the software project can be tested on the GitHub link provided in the footer.

Note!

All software projects are presented on <https://www.youtube.com/@ALTF4-26>

For Further Information:

Feel free to send an email to abdelrahmanelnakesh@gmail.com